EE3621: Digital Signal Processing

Lecture 16: Adaptive Filtering — LMS Algorithm

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Intro to Adaptive Filtering, non-stationary environments, and limits of static Wiener filters.
- **05:00 – 12:00 (7 mins):** Steepest Descent Algorithm, MSE cost function $J(\mathbf{w})$, exact gradient, eigenvalue spread.
- **12:00 – 20:00 (8 mins):** LMS Algorithm Derivation, instantaneous gradient $\hat{\nabla} J$, error $e[n]$, and weight update rule.
- **20:00 – 25:00 (5 mins):** Convergence conditions ($\mu < 1/\lambda_{\max}$) and time constant τ_k.
- **25:00 – 30:00 (5 mins):** Misadjustment M and Normalized LMS (NLMS).
- **30:00 – 33:00 (3 mins):** RLS Algorithm outline.
- **33:00 – 40:00 (7 mins):** Real-world Applications (Acoustic Echo Cancellation) and Checkpoint questions.

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2 Why Adaptive Filtering?

Classical digital signal processing often assumes that signals are stationary and their statistics (like autocorrelation) are perfectly known. If known, an optimal Wiener filter can be designed. However, in practice:

1. **Non-stationary environments:** Signal statistics vary over time.
2. **Unknown statistics:** We often lack prior knowledge of the channel or interference.
3. **Real-time constraints:** Computing and inverting large autocorrelation matrices in real-time is computationally unfeasible.

Adaptive filters dynamically adjust their coefficients to track changing environments. A standard FIR architecture is commonly utilized, as shown below for reference.

3 Steepest Descent Algorithm

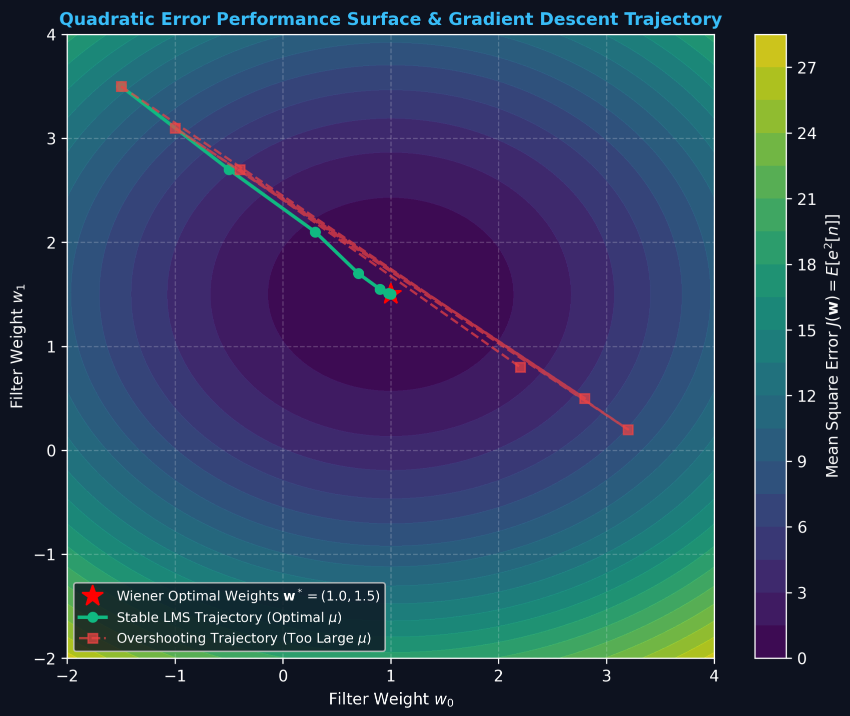

Figure 1: Quadratic MSE Error Performance Surface and Stochastic Gradient Descent Trajectory.

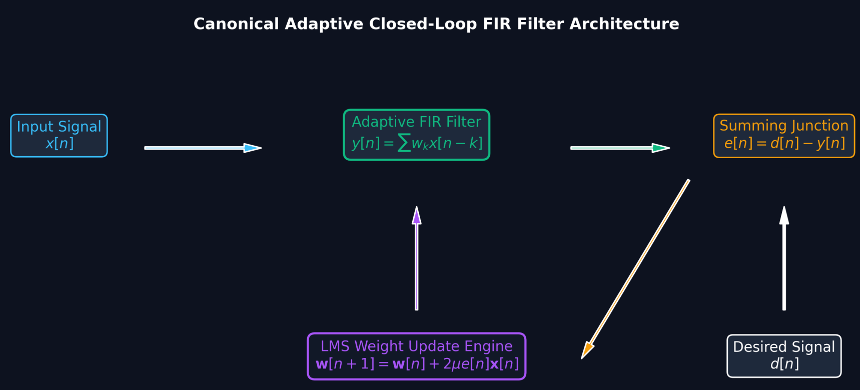

Figure 2: Canonical Adaptive Closed-Loop FIR Filter Architecture with Error Feedback Engine.

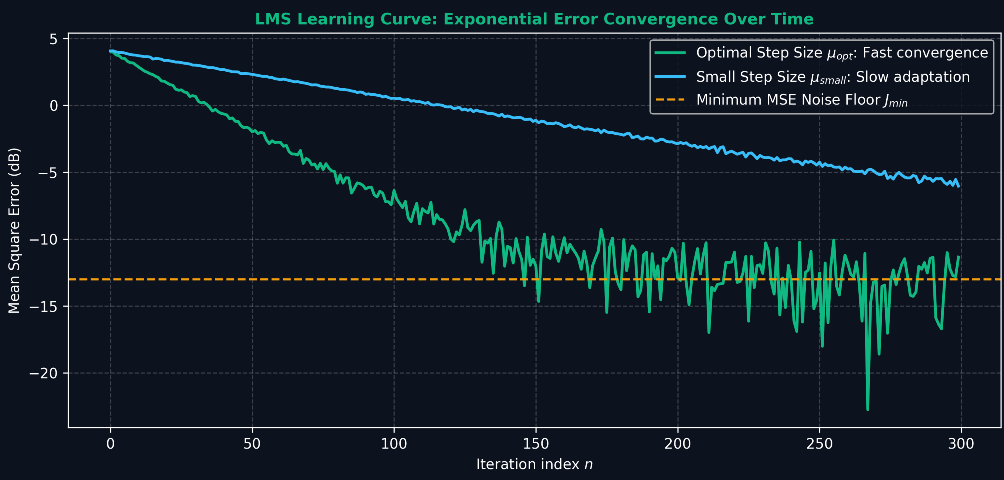

Figure 3: LMS Learning Curve: Exponential Error Convergence toward Theoretical Minimum MSE Floor.

Let the input vector be $\mathbf{x}(n) = [x[n], x[n-1], \ldots, x[n-N+1]]^T$ and the weights be $\mathbf{w} = [w_0, w_1, \ldots, w_{N-1}]^T$. The error is:

$$e[n] = d[n] - \mathbf{w}^T\mathbf{x}(n) \tag{1}$$

We minimize the Mean Squared Error (MSE):

$$J(\mathbf{w}) = E[e^2[n]] = E[d^2[n]] - 2\mathbf{w}^T\mathbf{r} + \mathbf{w}^T\mathbf{R}\mathbf{w} \tag{2}$$

where $\mathbf{R} = E[\mathbf{x}(n)\mathbf{x}^T(n)]$ and $\mathbf{r} = E[d[n]\mathbf{x}(n)]$. The minimum MSE is $\xi_{min} = J(\mathbf{w}_{opt})$ where $\mathbf{w}_{opt} = \mathbf{R}^{-1}\mathbf{r}$. We can rewrite $J(\mathbf{w})$ as:

$$J(\mathbf{w}) = \xi_{min} + (\mathbf{w} - \mathbf{w}_{opt})^T\mathbf{R}(\mathbf{w} - \mathbf{w}_{opt}) \tag{3}$$

The gradient is:

$$\nabla J = 2\mathbf{R}\mathbf{w} - 2\mathbf{r} \tag{4}$$

The steepest descent update rule is:

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \mu\nabla J = \mathbf{w}(n) + 2\mu(\mathbf{r} - \mathbf{R}\mathbf{w}(n)) \tag{5}$$

For stability, step size μ must satisfy:

$$0 < \mu < \frac{1}{\lambda_{max}} \tag{6}$$

4 LMS Algorithm Derivation

Since we don't know $\mathbf{R}$ and $\mathbf{r}$, we use the instantaneous gradient:

$$\hat{\nabla} J \approx -2e[n]\mathbf{x}(n) \tag{7}$$

Substituting this into the update rule gives the **LMS Algorithm**:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e[n]\mathbf{x}(n) \tag{8}$$

The convergence condition in practice is:

$$0 < \mu < \frac{1}{\mathrm{tr}(\mathbf{R})} = \frac{1}{NP_x} \tag{9}$$

The time constant for the k-th mode is $\tau_k \approx \frac{1}{4\mu\lambda_k}$.

5 Misadjustment and NLMS

The excess MSE due to gradient noise causes **Misadjustment** M:

$$M = \frac{\mu \text{tr}(\mathbf{R})}{1 - \mu \text{tr}(\mathbf{R})} \approx \mu N P_x \quad (10)$$

There is a trade-off: large μ yields fast convergence but high misadjustment.

To handle varying input power, the **Normalized LMS (NLMS)** adjusts μ dynamically:

$$\mu(n) = \frac{\alpha}{\epsilon + \|\mathbf{x}(n)\|^2} \quad (11)$$

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{2\alpha}{\epsilon + \|\mathbf{x}(n)\|^2} e[n]\mathbf{x}(n) \quad (12)$$

6 Recursive Least Squares (RLS)

RLS minimizes a deterministically weighted sum of squared errors with a forgetting factor λ:

$$\mathcal{E}(n) = \sum_{i=1}^{n} \lambda^{n-i} e^2[i] \quad (13)$$

It uses the Matrix Inversion Lemma to update $P(n) = \mathbf{R}^{-1}(n)$:

$$P(n) = \lambda^{-1}[P(n-1) - \mathbf{k}(n)\mathbf{x}^T(n)P(n-1)] \quad (14)$$

RLS has $O(N^2)$ complexity but much faster convergence.

7 Applications

Adaptive filters are used in acoustic echo cancellation, adaptive noise cancellation, channel equalization, and beamforming.

8 Key Formulas

Concept	**Formula**
MSE Cost Function	$J(\mathbf{w}) = \xi_{min} + (\mathbf{w} - \mathbf{w}_{opt})^T \mathbf{R}(\mathbf{w} - \mathbf{w}_{opt})$
Wiener Solution	$\mathbf{w}_{opt} = \mathbf{R}^{-1}\mathbf{r}$
LMS Update Rule	$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e[n]\mathbf{x}(n)$
Convergence Bound	$0 < \mu < \frac{1}{NP_x}$
Misadjustment	$M \approx \mu N P_x$
NLMS Update	$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{2\alpha}{\epsilon+\|\mathbf{x}(n)\|^2} e[n]\mathbf{x}(n)$

9 Checkpoints

- **Q1:** An LMS filter has length $N = 10$, input power $P_x = 0.5$W. Calculate max μ.
  Answer: $\text{tr}(\mathbf{R}) = 10 \times 0.5 = 5$. Thus $\mu < 1/5 = 0.2$.
- **Q2:** Let $\mathbf{w}(0) = [0, 0]^T$, $\mu = 0.1$, $\mathbf{x}(0) = [1, -2]^T$, $d[0] = 2$. Find $\mathbf{w}(1)$.
  Answer: $y[0] = 0$, $e[0] = 2$. $\mathbf{w}(1) = [0, 0]^T + 2(0.1)(2)[1, -2]^T = [0.4, -0.8]^T$.
- **Q3:** If μ is halved from 0.01 to 0.005 with $N = 20, P_x = 1$, what is the new M?
  Answer: $M \approx \mu N P_x = 0.005 \times 20 \times 1 = 0.1$. The convergence time τ doubles.